ONR FY2027 YIP: technical panel review

Trustworthy Collective Reasoning and Evaluation for AI Agents

Xinya Du | Mathematical Data Science, Code 311 | 28 October 2026

Scope and evidence. I read the complete technical narrative first: numbered pages 1-20, which are PDF physical pages 3-22 after the cover and contents. All proposal page locations below use the printed technical page numbers. The excluded material after numbered page 20 is not reviewed. Equations were checked against `main-full.tex`. This is a simulated three-perspective panel assessment, not three independent external reviews.

Version reviewed. Repository `xinyadu/YIP-Proposal`, main commit `029dff10d4eddbb9980dbd2d975dcb77756f35ed`; PDF Git blob `3e5ac71fb61be960cc2169cdcf723408d1d7417a`. The before/after comparison uses the repository's exact `white_paper_predrag-4page-v4.pdf`, not the different attached short white paper. The annotated FY27 NOFO governs compliance. The four funded examples are used only for calibration; the NSF files only for overlap. Their page numbers below are explicitly identified as PDF pages when necessary.

How to use this review. C3-C12 are the ten ranked funding problems. Later comments audit or operationalize them; they are not a second list of additions. Replacement passages are alternatives to the named original passages. Do not insert both a top-problem replacement and its corresponding line edit. No new experiments are requested beyond the five existing pilots. Future evaluation specifications below refine the already proposed research.

1. Panel verdict

C1. Decision: decline on the current technical proposal

Location: Sections 1.3 and 3.1-3.5, pp. 3, 5-18; eqs. (1), (5)-(6), Propositions 1-6.

Technical merit: Good. The proposal addresses a consequential problem and connects dependence-aware aggregation, impartial roles, evidence-checking debate, audits, and learning in a coherent architecture. Its principal weakness is soundness of the claimed connection between these components: the supplied propositions are largely inherited or elementary conditional facts, while the difficult guarantees for learned partitions, adaptive adversaries, reliable judges, and continued training remain unestablished. The fixed-total audit budget and continuing protection claim are incompatible as written, and the reinforcement-learning claims exceed Proposition 6.

Naval relevance: Good. The primary S&T Focus Area is explicitly and appropriately identified, and copied reports, compromised tools, scarce verification, and delayed truth give the research a plausible naval setting. The proposal does not yet tie its principal output to a sharply defined operationally meaningful error/cost tradeoff, and the example makes reproducibility of a port query appear to establish the truth of a live situation. The research remains broadly applicable basic research, but the naval case would be stronger with one bounded analytical use case and asymmetric consequences of error.

Mathematician/statistician: Technical merit **Fair**; naval relevance **Good**; **decline**. The model and decomposition are interpretable, but the apparent theorem count overstates the new mathematics. The missing truth orientation, dependence between training errors, and audit-budget scaling concern the assumptions on which the principal claims rest.

AI researcher: Technical merit **Good**; naval relevance **Good**; **borderline**. The failure modes and adversarial evaluation are well chosen. The missing comparisons to strong dependence-aware aggregation, tool-using single judges, recent tool debate, and peer-prediction training make it difficult to attribute a future gain to the proposed ideas, and message credit remains vulnerable to persuasion.

Navy S&T reviewer: Technical merit **Good**; naval relevance **Good**; **borderline**. The proposal names a valid focus area and potentially useful research products. The link from those products to fewer harmful assessment errors per verification budget needs to be explicit; the current autonomous patrol consequence and universal trust language make the capability claim less credible.

The one change that most raises the odds: rewrite Task 2.2 around a finite horizon T and an explicit audit budget/rate, replace the unsupported harm-rate promise with a research target H(T,B,Delta,alpha), and propagate that limitation to the overview. This removes a directly falsifiable inconsistency at the center of the promised safeguards. Correcting the font issue in C9 is separately non-negotiable for submission readiness.

C2. Three strengths to preserve

Location: Sections 1.1-1.2, pp. 1-3, Figure 1; task organization, pp. 7-16; Section 3.5, pp. 16-18.

- Keep the single central question and the progression from aggregation to debate to learning. Keep the running example, shortened and time-indexed.
- Keep the challenge / approach / preliminary results / evaluation and risks layout and the separate tasks. It makes the research auditable; a structural overhaul would waste the remaining day.
- Keep the red team's escalating capabilities, correct-minority versus overturned-majority metrics, and explicit failure regimes. These are more informative than another theorem with ideal assumptions.

2. Ten funding problems, ranked

C3. Rank 1 - a fixed total audit budget cannot supply the advertised continuing protection

Location: Section 3.1, pp. 5-6, R4; Task 2.2, pp. 13-14, Proposition 5; Table 4 G4, p. 18. **Must-fix.**

Reviewer: The proposal treats a fixed number of human audits as if it were a constant audit probability over an arbitrarily long stream. At audit rate rho, expected human work is rho*T. At fixed total B, rho must shrink with T; a one-time attack then escapes with probability approaching one. Not triggering audits based on suspicion does not make their total cost fixed. An anytime-valid false-flag test also does not itself prove attribution of collusion or a sublinear harm bound.

Replace the audit-budget and harm-promise sentences with:

> We analyze a finite stream of T decisions. Each is independently audited after commitment with probability rho, giving expected human cost B=rho*T. Proposition 5 concerns this audit rate; it does not imply continuing protection with a fixed total budget as T grows. We will characterize excess wrong decisions before exclusion as H(T,B,Delta,alpha), where Delta is an agent's excess error under randomized role assignments. Error flags identify unreliability, not intent. Sequential tests use predictions fixed from prior data; their guarantees require the stated conditional error model. We will measure detection and harm at each budget, including recognizable planted cases.

Cut/payment: replace the existing approximately 150-word stretch beginning "Flags lower the agent's weight" through "few roles," plus the tax-audit analogy (about 45 words), with this roughly 110-word passage. Use the saved space to define B, T and Delta beside Proposition 5 and change R4 to "Audit sampling is independent of suspicion; its expected cost is rho*T." In Table 4 replace G4's existing criterion, at comparable length, with "Harm and detection delay versus B and T; false-flag probability at most alpha under the stated null; no fixed-total-budget guarantee as T grows." These are substitutions, not extra paragraphs.

Also correct: "a weak record earns ... few roles" is false for the uniform judge lottery and can be false for singleton representative groups. An unflagged weak judge remains equally likely to judge. C7 and C16 specify the repair.

C4. Rank 2 - identifying latent components does not identify which answer is true

Location: Task 1.1, pp. 8-9, Proposition 2 and fitting paragraph; Sections 1.2-1.3, pp. 2-3. **Must-fix.**

Reviewer: Proposition 2 correctly says "up to relabeling," but the next paragraphs treat the resulting posterior as oriented toward truth. A population of agents with accuracy 0.8 and a population with accuracy 0.2 can induce the same unlabeled joint answer distribution after reversing the latent truth label. Arbitrarily coordinating agents need not follow the stationary latent-class model at all. The proposition therefore does not answer the headline problem for arbitrary unknown dishonest agents.

Replace the first sentences after Proposition 2 with:

> Proposition 2 identifies effective group response distributions, conditional on a fixed partition, only up to a permutation of truth labels. We orient and calibrate the posterior using

> separate development questions with known answers, excluded from fitting and policy updates. Answers alone cannot choose this orientation or handle arbitrary coordinated departures from the model. Our new statistical targets are partition recovery and calibration under specified separation, sample size, and contamination conditions; outside those conditions we report empirical performance without an identifiability guarantee.

Cut/payment: replace the existing opening "Proposition 2 states in advance ... adversarial" and the later sentence beginning "Because Proposition 2 fixes the partition" (together about 95 words). Keep a shortened one-sentence implementation statement about clustering and penalized likelihood, plus the already planned development calibration. The replacement is about 85 words. Replace "without human labels" in the overview with "without per-training-example truth labels, using separate calibration and audits"; pay its extra words by deleting the exam-grading analogy on p. 7.

Boundary to state: theory for stationary conditionally independent groups; adversarial robustness an additional research question, not a consequence of Proposition 2. Use time- or topic-stratified held-out data already in the plan; do not fit, orient, calibrate and evaluate on the same 300 pilot questions.

C5. Rank 3 - Proposition 6 and the actual rewards are being conflated

Location: Section 3.4, pp. 14-16; eqs. (5)-(6), Proposition 6; overview p. 3; pilot p. 15. **Must-fix.**

Reviewer: The proposition supports an unweighted conditional binary identity for a fixed reference independent of the scored answer. It does not prove truthfulness of the full chance-corrected objective, improvement under clipped/normalized GRPO, or resistance of message rewards to persuasion. Leaving out a group does not restore conditional independence after shared training or partition error. A same-policy cross-question subtraction is either a detached baseline, which does not change the expected scored-answer gradient, or part of an objective with additional gradient terms.

Replace the paragraph after Proposition 6 with:

> This is a conditional, per-question identity for a frozen reference and an action-independent weight. The cross-question term is treated as a detached baseline in the scored-answer update; it does not by itself change that update's incentives. An average reference accuracy above one half is insufficient if errors depend on the agent or reverse on a topic. Group exclusion removes direct self-influence under the fitted model, but does not establish independence under arbitrary collusion or shared training. We therefore freeze references during each update phase and monitor held-out truth accuracy and topic-specific reference errors.

Replace "When the judge decides by re-running evidence ... evidence shows" after eq. (6) with:

> Equation (6) is an experimental credit heuristic: prefix confidence can reward persuasive moves toward a wrong decision. Re-running tools verifies outputs, not the message's causal contribution or the truth of the final label. We test this reward against debate-win and terminal-decision rewards at matched compute.

Cut/payment: these two passages replace the approximately 160-word post-Proposition-6 paragraph and the approximately 40-word persuasion claim. The shorter replacement funds

"frozen reference" and "detached baseline" in the proposition itself. Delete the pilot's comparison of final accuracy gain to 1-2*eta; that prediction does not follow from the proposition. Replace it with measured held-out learning curves if the pilot runs. Do not substitute invented numbers.

One-day scope: narrow the guarantee; retain message credit as a falsifiable hypothesis. Do not attempt to invent a new convergence or incentive theorem before submission.

C6. Rank 4 - the advertised rater-model novelty is too close to an existing model

Location: Sections 1.3 and 2, pp. 3-5; Task 1.1, pp. 7-9, eq. (2); Table 1, p. 6. **Must-fix.**

Reviewer: Jaffe et al. (2016), already cited generically, uses the same binary truth -> independent latent group responses -> noisy member outputs hierarchy. Equation (2)'s binary version is a symmetric latent-error special case. This is more than distant inspiration. The new contribution must be distinguished from adaptation, and the proposed experiments must include appropriate comparators.

Replace "We propose a shared-factor rater model ..." on p. 7 with:

> We adapt the dependent-classifier hierarchy of Jaffe et al. (2016) to debate routing. The new questions are when learned groups yield calibrated decisions, when spending on a debate reduces expected loss, and how those properties degrade under strategic contamination.

Replace p. 3's "Why it will succeed" opening with:

> Established results justify the idealized design choices; the proposed research studies the finite-sample and adversarial conditions under which they remain useful. The main mathematical targets are calibration after learning groups, quality versus impartiality, and excess harm versus audit budget.

Cut/payment: the first replaces the rater-model assertion plus exam analogy at similar length. The second replaces the existing claim that each method follows from a property "we prove" and shortens the repeated list of prior systems; no new paragraph is needed. Credit the model directly next to eq. (2), not solely in related work. Recast novelty item (i) as these new questions; preserve the length of the existing item.

Closest work already in the bibliography: dependence-aware Ising aggregation, CARE, Tool-MAD, DynaDebate, and peer-prediction training. C8 gives the minimum comparison plan. I did not verify that no prior system ever re-executes a debater's exact call; replace that universal negative with a description of the reviewed comparators' designs.

C7. Rank 5 - independence and reproducible calls are not a reliability certificate for the judge

Location: Task 1.2, pp. 9-11, eq. (4), Proposition 3; Task 2.1, pp. 11-12, Proposition 4; Figure 1, p. 2. **Must-fix.**

Reviewer: The judge is uniformly random among held-out agents, irrespective of demonstrated step competence. A different model may still be inaccurate or dishonest. Re-running a stale or compromised database returns the same wrong answer, and the same true record can support an invalid inference. Proposition 4 assumes away these problems with a perfect single-step judge. The protocol also lacks an explicit result when both arguments lose, both survive, or the correct answer lacks an honest advocate.

Replace the last sentence before Task 2.1 and the tool-error-rate assertion on p. 12 with:

> Impartiality limits manipulation of selection, not judging error. Before a run, the judge pool is fixed using held-out step checks for the relevant tool and topic; the lottery is uniform within that eligible pool. Re-execution checks authenticity of an output. Correctness also requires an appropriate query, current and trustworthy source data, and a valid inference from result to claim. If neither side is supported, both fail, or no eligible judge is available, the output is unsettled and excluded from training. We measure coverage as well as error.

Cut/payment: replace the roughly 40-word "judge need not be wiser" sentence and the roughly 55-word Verification ending "tool's error rate," plus shorten the generic definition of a harness by about 15 words. The replacement is about 95 words. Eligibility uses the existing development evaluation, not a sixth pilot or triggered human review. Replace the final Task 1.2 risk sentence with "If no eligible model-disjoint judge exists, mark the item unsettled; varied prompts do not establish independence" at comparable length.

Keep the assumption visible: a correct side must actually surface and dispute the false step within the allotted exchange. A role instruction does not ensure this, especially when all initial answers agree or a selected debater is compromised. Qualification of this claim is more useful than a longer proof of Proposition 4.

C8. Rank 6 - the evaluation can credit tool access and weak baselines to the new method

Location: Task evaluations, pp. 9-16; Section 3.5, pp. 16-17; Tables 2 and 4, pp. 16 and 18. **Must-fix.**

Reviewer: The main listed baselines omit the closest methods discussed in related work. A debate that can query the decisive database should beat a text-only judge even if debate adds nothing. "Calibration," "preset bound," and "more accurate than all baselines" do not define an executable success criterion; OpenReview agreement or a reproducibility report does not give a truth label for every scientific claim.

Replace the existing metrics/baselines paragraph with:

> Report decision error including abstentions, selective error and coverage, Brier score, and fixed-bin calibration error. Report minority recovery conditional on an initially correct minority, majority damage conditional on an initially correct majority, excess wrong decisions per 1,000 cases, audits-to-flag, and false-flag probability. Match total inference tokens and tool-call budgets. Compare aggregation with the dependent-classifier model and CARE;

> debate with a tool-using single judge and tool debate; learning with peer-prediction and cohort rewards. Use paired question resampling and separate seeds. In C, score only independently verified executable or planted claims; review scores are preferences, not truth.

Cut/payment: this roughly 110-word replacement takes the current approximately 100-word metrics/baselines/statistics paragraph plus the generic "Each provides known answers" phrase in Table 2's caption. Change the corresponding Table 2 C row at equal length to "Verified executable claims in public papers; planted errors; review scores reported separately as preference agreement." Do not claim all papers have known answers.

Table 4 substitutions, not additions: G1: "Paired improvement over dependence-aware baselines at matched budget; calibrated held-out posteriors; seeded-group recovery." G2: "Zero within-group rating manipulation; coalition gain versus size and eligible groups; quality ratio on a fixed nonnegative scale." G3: "Paired error reduction versus a tool-using judge and tool debate at matched budget, with coverage reported." G4: use C3. G5: "Held-out learning gain over peer/cohort and debate-win rewards; report attack success and shared-error change." Replace the existing cells, which are longer. Use 95% intervals as the common reporting convention, not a manufactured numerical target.

Delete the unsupported H1 prediction: fewer than three independent groups removes the stated identifiability guarantee; it does not imply that every empirical advantage over voting disappears. No additional predeadline experiments are needed to correct the planned comparisons.

C9. Rank 7 - the current typography is a concrete compliance risk

Location: Figures 1-5, pp. 2, 8, 10, 11, 13; Tables 1-4, pp. 6, 16, 17, 18; captions throughout. **Must-fix.**

Reviewer: NOFO Section D, printed pp. 15-16, specifies Times New Roman, 12-point, with no stated small-font exception for figures, tables or captions. This source deliberately uses scriptsize (about 8 points) inside figures, small (11-point nominal) tables/captions, and `mathptmx`, which embeds a Times-compatible Nimbus Roman face rather than Times New Roman. The 20-page claim currently depends partly on small text.

Change: set ordinary figure/table/caption text to 12-point; use the required face or obtain grants-office confirmation of an accepted equivalent. Do not treat ordinary 12-point TeX text extracting as 11.96 PDF points as a size violation: that is unit conversion. The actual scriptsize/small settings are the problem. Mathematical subscripts are a separate typographic issue, not evidence that all math is noncompliant.

Cut/payment: delete Table 1 outright (roughly half a page or more) and remove its two callouts; Section 2 already supplies its content. Delete Figure 5 (roughly a quarter page), whose hypothetical 5%/40% bars repeat the audit explanation. Rebuild Figure 1 with brief 12-point entries instead of four paragraphs; retain the five-agent input, three stages, one disputed step and one limitation, cutting at least half its words. Compress Figures 3-4 to labels and actions already explained in text. Reserve these recovered areas for font expansion; do not also spend them on new prose. Recompile and recount before considering optional edits.

Reason to rank this separately: it can defeat an otherwise acceptable submission. Older funded examples do not establish a FY27 exception. No assurance of ONR rejection or acceptance can be inferred from the PDF alone.

C10. Rank 8 - expected pilots currently look like completed evidence

Location: preliminary results in Tasks 1.1, 1.2, 2.1, 2.2 and 3, pp. 9, 10, 12, 14, 15; Figure 2(a), p. 8.
Must-fix.

Reviewer: All five orange passages contain precise past-tense numerical results that have not been measured. They cannot support feasibility or a proposition. The uncolored simulation also needs a defensible provenance: the inspected repository stores plot coordinates in the TeX, but I did not locate its generating script/data. That is an inability to verify the result, not evidence it is fabricated.

Change: replace each unrun pilot in full with its prospective sentence in C28-C32; remove all imagined estimates. For completed pilots, state actual model IDs, sample split, number of independent runs, uncertainty and the comparator using the same available resources. Replace the theory-to-final-accuracy prediction in Task 3 even if the pilot runs.

Cut/payment: each prospective replacement is shorter than the orange passage it replaces. If Figure 2 is measured, use the caption's existing space to identify disjoint fit/test questions, how labels were oriented, how calibration error was computed, and which 60 runs were inspected. Cut the p. 7 verbal retelling of the same curves. If those details cannot be substantiated, label the figure "illustrative model simulation" and remove the run-count, recovery, standard-error and calibration claims; do not let a qualitative illustration impersonate a measured study.

Final-build check: replace/remove every `pending` passage and switch `finalbuildtrue`. The build guard detects pending macros, not unsupported numbers elsewhere.

C11. Rank 9 - the overlap boundary is asserted rather than drawn

Location: Section 4 "Relation to other support," p. 19; Tasks 2.1 and 3, pp. 12 and 15; testbeds B/C, p. 16.
Must-fix.

Reviewer: Reusing an established QA pipeline is legitimate, but the pending hypothesis proposal also develops an execution sandbox, reproducibility evaluation, fine-grained rewards and agent RL. Those are close to parts of this proposal. The complex-reasoning proposal already proposes event-based QA generation and explanation evaluators. "Any overlap ... would be removed" does not tell the reviewer which work this award buys. The named Open Philanthropy debate-control award may be even closer, but its statement of work is not supplied here.

Replace the NSF portion with:

> CAREER supports event extraction and graph construction; this project reuses those outputs rather than redeveloping that pipeline. The pending NSF reasoning proposal studies explanation generation and evaluation; the hypothesis proposal studies discovery and execution-based implementation training. ONR supports dependence-aware collective decisions, impartial roles, adversarial audits, and multi-agent credit under noisy references. Shared infrastructure and prior results will be identified separately from ONR-supported aims and effort.

Cut/payment: replace the current approximately 70-word support paragraph and cut about 15 words from the repeated award/service list in Section 5. Retain a compact, verified sentence distinguishing the Open Philanthropy and Amazon award aims and charged effort; I cannot supply that factual distinction without their award scopes. Use the generic service list, not research-method space, if that sentence needs another line.

Action within a day: check the actual scopes with the current/pending-support records and make the description consistent. Do not assert that another award excludes an aim merely because this narrative says it does. Detailed NSF comparisons appear in C41-C43.

C12. Rank 10 - the naval example turns an assessment aid into an unjustified operational decision

Location: Section 1.1, p. 1; Figure 1, p. 2; Section 1.4, pp. 3-4; eq. (3), p. 9. **Should-fix.**

Reviewer: The copied-report problem is plausible, but five AI answers do not establish that a patrol should launch, and a 09:10 port record does not answer an unspecified present-time question. The list of maintenance, logistics, cyber and autonomy uses broadens the claims faster than the plan supports. The strongest Navy connection is an analyst's source-backed assessment under scarce verification, not a claim of deployment readiness.

Replace the opening example's operational framing with:

> Consider an illustrative assessment at a maritime operations center: at 09:10, has vessel V departed? Five AI tools consult schedules, port records and tracks. Three report departure, but two copy an outdated schedule and one is compromised. The officer needs a time-indexed assessment and the evidence supporting it before deciding whether further verification or patrol tasking is warranted. A three-to-two vote treats copied information as confirmation; debate is useful only if it exposes a check the officer could not obtain as cheaply from one tool.

Cut/payment: this replaces the existing approximately 140-word example with about 95 words. In Figure 1 replace "Decision: No ... no patrol is sent" with "Assessment at 09:10: no departure supported; officer retains tasking authority," at similar length. Replace the maintenance/logistics/cyber list bullet and the next generic task-mapping sentence with: "The naval evaluation asks how many false departure assessments and missed departures remain at a fixed verification budget as reports become stale, copied or forged. These mechanisms apply to any source-dependent assessment stream." This is shorter than the cut.

Clarify eq. (3): L is a symmetric toy loss. For this naval use case, specify separate false-positive and false-negative costs in the planned evaluation; replace "so higher stakes lead to more debates" with "This symmetric-loss illustration will be extended to distinct costs for missed and false departure assessments." No new experiment is required.

3. Mathematical audit

C13. Equation (1): valid conditional bookkeeping, not an end-to-end guarantee

Location: Section 3.1, pp. 6-7, eq. (1).

Correct? Yes as a union bound, if the epsilons are conditional probabilities or valid upper bounds in the routed-debate population and the leading-answer probabilities are calibrated on the deployment distribution. Explicitly let B be the debate event, C the event that neither candidate is correct, S successful adversarial steering, and E a remaining judging failure. For example, use eps_cand=P(C|B), eps_role=P(S|B,C^c), eps_step=P(error|B,C^c,S^c). The union bound then gives the displayed inequality. Ordinary calibration on an honest development set does not establish the assumption after adversarial routing or RL.

Counterexample to the unconditional claim: let every agent return the wrong answer, the fitted posterior assign it 0.99, tau=0.95, and no item go to debate. Actual error is one; the first term is 0.05 and the debate term is zero. This violates calibration, so it refutes the interpretation as an unconditional safeguard, not the conditional inequality.

Novelty/credit: law of total probability and a union bound; no special source is needed and it should not be sold as new theory. **Design justified:** routing, candidate coverage, role integrity and local judging can be diagnosed separately. **Load-bearing?** Useful organizing device; decorative if called a proof that the integrated system works. The quantities are currently not computable guarantees for an untrusted pool.

Minimal change/payment: replace "Removing a task leaves its term uncontrolled" and the preceding Task 3 monotonicity claim with "The epsilons are conditional on routing and the preceding failures being absent; (1) organizes error accounting, not an unconditional trust guarantee. Task 3 may change every term in either direction." Same approximate length. In experiments distinguish corrupted-role occupancy from successful steering: the latter requires a defined matched attack intervention or an upper bound; it is not observable from an ordinary transcript alone. Pay that metric definition by shortening the repeated results list on p. 17.

C14. Proposition 1: correct saturation under a specified channel

Location: Task 1.1, p. 8, eq. (2), Proposition 1.

Correct? The binary likelihood ratio calculation is correct for 0<lambda_g<=1/2 and observations with defined likelihood. More precisely, the absolute log-likelihood contribution is bounded by log((1-lambda_g)/lambda_g); at lambda_g=0 the extended bound is infinite and says nothing. With symmetric K-class wrong answers, the stated pairwise bound is valid when the truth component is at least as likely as each wrong component (lambda_g<=(K-1)/K). The binary restriction already suffices if retained.

Realism: all dependence must pass through a fixed group factor. Overlapping sources, question-dependent groups, common difficulty and adaptive coordination can violate that representation. One can force a cap by modeling choice; the proposition does not establish that the fitted cap equals the information content of real agents.

Novelty/credit: elementary channel likelihood-ratio bound inside the hierarchy; directly credit Jaffe et al. as in C6. I did not verify a prior publication of this exact wording. **Design justified:**

saturating duplicated evidence rather than multiplying copies. **Load-bearing?** Yes for an ideal-model design property, not evidence of a new statistical theory.

Important identification gap: effective group distributions can be identifiable while lambda and member error are not separately identifiable. For a singleton binary group, observed error e=lambda+mu-2*lambda*mu. e=0.2 is consistent with (lambda,mu)=(0.2,0) or (0.1,0.125), both with invertible member channels. Their claimed latent caps differ. Proposition 2 does not identify the decomposition needed to interpret lambda as a recovered source reliability. Keep claims about observable/effective likelihoods separate. Add this limitation by replacing the sentence beginning "So a group carries ..." with a shorter statement of the effective cap plus this caveat; do not add a new proof.

C15. Proposition 2: the theorem application is sound; the operational reading is not

Location: Task 1.1, pp. 8-9, Proposition 2; eq. (2).

Correct? With K>=2 truth labels, positive mixing weights, a common fixed response model, three jointly observed independent views whose component-response matrices have full column rank K, the Kruskal condition is 3K>=2K+2. Thus effective distributions are identifiable up to label permutation. For incomplete answer panels, merely having some three groups per item is insufficient unless the observation pattern links the views and provides the needed joint distributions; specify a common overlapping design. The two-view nonidentifiability statement is generally correct.

Counterexample to truth recovery: Y is balanced binary and three groups are independent with accuracy 0.8. Define Y'=1-Y and retain the same observed answers. The same unlabeled distribution is now generated by groups with accuracy 0.2. Full rank and positive priors hold in both cases. Neither three views nor invertible member confusion matrices chooses the truth orientation. This is already allowed by "up to relabeling"; it is the surrounding inference that is false.

Realism: stationary independent groups and full rank are strong assumptions for LLMs; role prompts do not certify them. Dishonesty must fit a specified contamination model or fall outside the theorem. **Novelty/credit:** a credited application of Kruskal (1977)90069-6) and Allman et al. (2009). **Design justified:** sufficient independent views and a separate orientation resource. **Load-bearing?** The most relevant existing result in the draft, but it does not solve learned partition recovery, finite-sample calibration or arbitrary collusion. Apply C4; retain the proposition.

C16. Proposition 3: correct rating invariance; a very weak quality guarantee

Location: Task 1.2, p. 10, eq. (4), Proposition 3; lottery and risks pp. 10-11.

Correct? Part (a) holds conditional on first answers, the partition, eligibility, historical weights, and tie-breaking being fixed independently of current ratings. Part (b) follows immediately if the weighted rating rule actually selects the highest-quality eligible member of every drawn group. "Honest ratings" alone does not imply this without a specified aggregation/ranking assumption; the text does supply that ranking assumption. If the best agent is excluded by the judge's group, pi*=0 and the bound is vacuous.

Scope counterexamples: with three eligible singleton groups supporting yes and one supporting no, a yes agent can switch its first answer to no and raise its representative probability from 1/3 to 1/2, despite invariant ratings. With ten singleton groups of qualities (1,0,...,0), a uniform group draw yields expected quality 0.1. With one excellent held-out judge and nine incompetent ones, uniform judging selects the excellent one only 10% of the time. These do not contradict the stated proposition; they contradict broad readings of "impartial" as reliable or fully strategyproof.

Novelty/credit: partition-based impartiality is credited in the text; Alon et al. (2011) and Kurokawa et al. (2015) are the relevant lineage. I could not access the latter's full text through the web tool, so I do not certify the claimed superiority over its 1/(1+m) bound. Different feasible sets, exclusions and quality definitions already prevent the stated numerical comparison from establishing dominance.

Design justified: remove within-group influence via current ratings. **Load-bearing?** The invariance is; the quality bound is essentially "we pick the best group with probability pi*" and is not a strong optimality result. Replace the sentence beginning "with two or three groups ... this beats" with "The quality bound can be small or zero; we compare mechanisms under the same eligibility constraints and utility scale." This shorter replacement pays for the fixed-profile qualifications. Retain cross-group coalition gains as proposed research.

C17. Proposition 4: a correct idealized implication that assumes away the hard judging problem

Location: Task 2.1, pp. 11-12, Figure 4, Proposition 4.

Correct? Yes if an honest advocate for the correct candidate exists, actually exposes the false step within the allowed protocol, and the judge evaluates truth/entailment correctly with sufficient context. The proof is a direct implication of its assumptions, not an efficiency or robustness theorem. Say "disputes a false step" rather than only "can find" it if asserting a guarantee of the actual run. Otherwise state existence of a winning strategy.

Counterexample to the harness claim: a stale database says "at berth" at 09:10; both the original call and its rerun return that record. The vessel leaves at 09:20 and the question is asked at 10:00. Reproduction is perfect, yet the inference "still in port now" is false. Likewise true quoted premises can be selectively incomplete. These cases violate the perfect-judging/valid-argument assumptions; they show that tool replay does not enforce them.

Novelty/credit: inherited debate principle, already credited to Irving et al. (2018) and Brown-Cohen et al. (2024). Their formal results should not be equated with this informal numbered-step protocol. **Design justified:** local disputes and explicit losing rules. **Load-bearing?** An interface specification; currently decorative as mathematical justification for real weaker judges. C7's reliability conditions, unresolved outcomes and measured error types make it useful. Replace the existing perfect-tool conclusion; do not expand the proof.

C18. Proposition 5: correct Bernoulli exposure fact; not a detection or budget theorem

Location: Task 2.2, p. 13, Proposition 5; Table 4 G4, p. 18.

Correct? For a specified n committed attack attempts and independent post-action Bernoulli draws, the chance of zero audit exposure is (1-rho)^n. Adaptivity to past information does not reveal the next independent draw. Keep the distinction between exposure, correct audit labeling, attribution and flagging. If the number of attempts depends on detection/stopping, do not condition on eventual attack count and reuse the formula without stating the stopping convention.

Counterexample to the budget interpretation: audit B=10 uniformly chosen decisions out of T=10,000. A single attack is inspected with probability 0.001. Increasing T tenfold with B fixed reduces that to 0.0001. For a fixed attack set, zero-exposure probability with exactly B audits is choose(T-n,B)/choose(T,B), not the Bernoulli formula. Even with perfect labels, expected attacks before exposure at constant rho is 1/rho; that cost scales with T/B at a fixed total budget.

Realism: hiding a post-commitment random draw is plausible. Having reliable truth for every sampled decision is not assured in testbed C or superhuman oversight, and attackers can hide traces needed for attribution. **Novelty/credit:** elementary independent sampling; the nearby Steinhardt et al. (2016) result concerns batch crowdsourcing/curation with trusted ratings, not this online harm claim. **Design justified:** unpredictable real audits supplement recognizable plants. **Load-bearing?** Only after C3 aligns the resource model and limits the conclusion.

C19. Proposition 6: correct conditional identity, but insufficient and ambiguous for eq. (5)

Location: Task 3, pp. 14-15, Proposition 6, eqs. (5)-(6), GRPO pilot.

Correct core: for fixed q,y,x and a frozen binary reference with conditional error eta_q independent of the scored action, `E[1{x=D}|q,y,x] = eta_q + (1-2*eta_q) 1{x=y}`. The scored-action REINFORCE gradient is scaled by 1-2*eta_q if weights are fixed independently of that action. This is an elementary noisy-label calculation. State the conditioning; the current display suppresses it. With class-dependent noise, eta needs its appropriate conditioning. With nonuniform multiclass errors, the scalar formula stated in the following paragraph does not hold without symmetric wrong labels.

Counterexample 1 - shared mistakes despite a good marginal reference: let X=D and both be wrong on exactly 40% of questions. Marginal reference accuracy is 0.6, yet copying D achieves agreement one, while always giving the truth agrees only 0.6. Leaving out a group prevents literal self-inclusion but not this correlation across groups or after training.

Counterexample 2 - baseline versus full objective: balanced implementation details matter even without noisy labels. Let P(Y=1)=0.9, D=Y, and an input-ignoring same-policy report be Bernoulli(p). Same-question expected agreement is 0.1+0.8p; cross-question expected agreement is also 0.1+0.8p. The full expected score in (5) is identically zero with derivative zero, whereas the truth-agreement objective has derivative 0.8. Thus full-objective gradient equivalence is false. If the second term is detached and only the scored answer receives an update, it is a baseline and the conditional gradient identity holds; then subtracting it does not

independently change the expected update's incentives. Choose and describe the implementation.

Other limits: per-question positive scalings need not preserve the direction of the average shared-parameter gradient. With equally likely questions whose true-reward gradients are +1 and -2 (locally, correct-answer probabilities 0.5+theta and 0.5-2*theta) and reference error rates 0.01 and 0.49, the true average is -0.5, while the noisy average is (0.98-0.04)/2=+0.47. Posterior-dependent omega also requires a weighted-noise analysis; eta alone need not describe it. None of these identities predicts a 73% fraction of final accuracy improvement, and GRPO clipping, normalization and KL regularization are outside the proof.

Novelty/credit: chance correction is credited to Dasgupta and Ghosh (2013) and Shnayder et al. (2016); the affine identity is elementary, not a new training guarantee. **Design justified:** frozen independent references, explicit noise regime, and cautious confidence weighting. **Load-bearing?** Potentially, if C5 restricts the claim; presently overextended. It does not justify eq. (6), which has no truth-gradient result at all.

4. Does the full proposal pass the "ad hoc" test?

C20. Task 1.1 - substantially improved, but the new mathematical problem must be separated from the inherited model

Location: pp. 7-9, eqs. (2)-(3), Propositions 1-2; white paper pp. 2-3.

The white paper proposed unusual-agreement clustering and shared factors. The full draft now gives a generative likelihood, a saturation property, a conditional identifiability result and an expected-loss routing rule. This is the clearest improvement in mathematical reasoning. What remains heuristic is how partitions, topic effects and adversarial priors are estimated and validated together; treating topic/difficulty terms as an automatic cure can itself add latent confounding. Make the stated research target a finite-sample calibration/routing guarantee under a specified contamination and group-separation regime. That can be a proposed theorem, not a claim already proved. Use C4/C6's replacements; pay for the target by cutting the existing promise to "characterize how many questions" and restating it precisely at the same length.

Equation (3) is a valid one-step value-of-information rule only with calibrated pi and an applicable conditional debate error estimate. If debate error varies with pi, topic or adversary, write epsilon_D(q) and c(q), or explicitly call the constant-error rule a development-set approximation. At equal length replace "worth its cost exactly when" with "under this constant-error approximation, worth its cost when." This is more defensible than claiming disagreement/confidence rules can never route unanimity: confidence-based routing can also route low-confidence unanimous cases; the distinction is a dependence-corrected confidence estimate.

C21. Task 1.2 - derived from impartiality, but not from reliable-judge selection

Location: pp. 9-11, eq. (4), Proposition 3, Figure 3; white paper p. 3.

The full draft specifies the lottery and excludes the judge from rating, which closes a white-paper ambiguity. Proposition 3 explains a particular manipulation restriction. It does not derive the chosen quality lottery as optimal, protect strategic first answers, or prevent cross-group collusion. With singleton groups, peer ratings do no selection work at all. State the conditional invariance and use the already planned quality/manipulability evaluation to characterize the tradeoff under a fixed eligibility set. Replace the cross-paper dominance sentence with C16's qualification. The "few models -> varied instructions" fallback is a heuristic; C7's unresolved outcome is a mathematically honest boundary.

C22. Task 2.1 - principled failure analysis, still an empirical protocol

Location: pp. 11-12, Proposition 4, Figure 4; white paper p. 3.

The numbered-step losing rule, candidate false-step search, context reply and recursive refinement improve on the white paper's loosely specified stopping point. The perfect-judge proposition supplies an ideal contract, not a derivation of a useful finite-budget implementation. A principled version identifies three separately scored failures: challenger misses the false step; tool/query/source is unfit; judge misreads a valid observation. Then the pilot measures which part replay fixes. Apply C7 within the existing Verification paragraph; use its cuts. The actual method can remain experimental without being described as a theorem-backed guarantee of trustworthy debate.

C23. Task 2.2 - principled random sampling, heuristic blame and unsupported rates

Location: pp. 13-14, Proposition 5, Figure 5; white paper pp. 3-4.

Unpredictable real sampling directly addresses evaluation awareness. The full draft adds sequential testing and a detection/harm ambition, but it does not define the null predictive model, minimum excess error, per-agent exposure or the budget scaling. Comparing agents across pairings does not automatically disentangle collusion from honest difficulty or a bad partner. A principled task specifies the data-generating/null assumptions and targets harm versus audit cost with a lower bound or impossibility regime. C3 provides this in fewer words than the current claim. Controlled-edit plants, discriminator training and paraphrasing remain plausible interventions; do not present them as guaranteed indistinguishability or collusion removal.

C24. Task 3 - partly principled first-answer score; message credit remains a heuristic

Location: pp. 14-16, eqs. (5)-(6), Proposition 6; white paper pp. 3-4.

Adding first-answer rewards, exclusion of the agent's group and an explicit noisy-reference condition is a substantial change from the white paper's message-only training. The first-answer result is useful within its narrow regime. The message score is still outcome-conditioned confidence change; its telescoping sum does not make it truthful, causal or incentive compatible. The draft correctly says policy invariance does not carry over from Ng et al. (1999); keep that caveat. A principled presentation makes eq. (6) a candidate to be falsified, freezes the evaluator within an update phase, and separates its evidence/provenance error from persuasion. Apply C5; cut the unearned claim that replay makes the decision independent of persuasion.

C25. Colleague's questions: current answer status

Location: Tasks 1.1-3, pp. 8-17.

Question	Status	Where and what is still missing
(a) Reliability without truth, with correlation and dishonesty	Partly answered	Proposition 2, p. 8, supplies rank and conditional-independence assumptions for fixed groups. Truth orientation, learned grouping and arbitrary dishonesty remain outside it; audit flags alone do not repair identifiability.
(b) Why trust the judge?	Partly answered	pp. 10-12 provide independence of selection, replay and later audit. They do not qualify step competence or source truth; Proposition 4 assumes correctness.
(c) Wrong group decisions reinforcing behavior	Partly answered	pp. 14-16 state eta<1/2 and a stop rule. They do not establish conditional independence, calibrated topic-level error, or the message-reward condition. Average held-out accuracy can hide a badly wrong subpopulation.
(d) Concrete metrics	Answered at the metric-family level; partly specified operationally	Task evaluations and p. 17 list meaningful metrics. Definitions, denominators, abstention coverage and strong comparison plans still need C8.

This table is an audit result, not proposed text to insert; it consumes no proposal pages.

C26. The program officer's six questions

Location: Sections 1-3.7, pp. 1-19.

PO question	Assessment of this draft
What problem?	Clear on p. 1: reliable collective assessment with unknown, dependent and potentially strategic agents.
Why important?	Clear on pp. 1-4; several empirical failure claims and a plausible naval assessment setting.
How solved today and limits?	Strong coverage on pp. 4-5, but the relationship to the closest models is understated and the category table overclaims.
How solve, what new/better, why work?	Architecture clear on pp. 2-3; novelty and mathematical sufficiency remain the main weaknesses on pp. 7-15.
What limitations?	Present in every task and p. 18, but incompatible stronger claims in the overview prevent these caveats from controlling the reader's conclusion.
What impact?	Stated on pp. 3-4 and 17-18; needs the budgeted error tradeoff rather than an implication of label-free, fixed-human-cost trust at unlimited scale.

The full proposal answers the six questions in form. It does not yet answer the mathematical objection in substance for the adversarial, learning setting. No rearrangement of sections is needed.

C27. White-paper changes and possible dropped commitments

Location: white paper pp. 2-4 versus full proposal pp. 6-16 and 18.

- **Preserved:** central question, three RGs/five tasks, independent initial answers, copied-source example, role impartiality, optional tools with exact-call replay, planted plus random real audits, three testbeds and trained collusion attacks. No major approved task appears to have been dropped.
- **New:** dependence likelihood and propositions, cost/stakes routing including unanimous cases, judge-first draw with no ratings by the judge, explicit step losing rule, context reply/recursive checks, sequential tests, first-answer reward and group exclusion, formalized message credit, schedule and failure regimes.
- **Changed:** the white paper trained on the group's final decision with tool-backed decisions weighted more. Eq. (5) now uses a leave-group-out aggregation reference, not the final debated decision. State this distinction in Figure 1/overview; replace "RG3 trains every agent on the decision" with "RG3 trains first answers against held-out-group references and messages against the audited debate outcome." Pay its extra words by deleting the redundant feedback-arrow label. This is a substantive change the PO should be able to see.

- **Changed/qualified:** the white paper's simple "stop at the first differing step" becomes a short exchange, recursion and sometimes full-argument judging with an unsettled result. Define how these outcomes are counted; do not pretend they all meet Proposition 4.
- **Omitted detail worth restoring only if substantiated:** the white paper explicitly says tool-backed decisions weigh more; the full proposal uses judge-confidence weights, which is not the same rule. Prefer a sentence saying which weight is actually implemented rather than adding both. Replace the existing omega-definition explanation at equal length. Its cited 2.3% unnecessary-search result also disappears; this is not a broken commitment and does not warrant restoring another number.
- **Not verified:** that the PO endorsed each individual method or empirical claim. Approval of the white paper and structure is not approval of the additional guarantees or these technical changes.

6. The five orange pilots

C28. Pilot 1: shared-factor aggregation - run this first

Location: Task 1.1 preliminary results, p. 9; Figure 2, p. 8.

Would the expected result convince me? It would support feasibility under an intentionally seeded dependence pattern, not natural group recovery or adversarial robustness. A 0.87 versus 0.74 gap matters only with disjoint unsupervised fitting/development/testing, an explicit label-orientation method, and the same initial agent answers for all aggregators. "9/10 recovered" needs an exact group-recovery definition. Recovery from a shared hint is a favorable test of the assumed model.

Read out from this same pilot: actual accuracy difference and uncertainty, held-out Brier/calibration, group recovery, and how models behave without the flawed hint. Do not claim a state-of-the-art advantage if only the independent rater comparator was implemented. No extra pilot is requested.

If unrun, replace the entire orange passage with: "Planned pilot: nine agents on 300 GSM8K problems will test whether shared flawed hints produce recoverable dependence and whether dependence-aware aggregation preserves accuracy and calibration relative to voting and the independent rater model." This is shorter than the original. Retain only the independently documented prior PRD evidence as completed work.

C29. Pilot 2: impartial roles - sanity check, not the strongest evidence

Location: Task 1.2 preliminary results, p. 10.

Would the expected result convince me? Zero selection gain from the specified current-rating attack should follow from the mechanism and mainly checks implementation. The useful result is the quality cost (the proposed 0.92), but "best arguer's quality" needs a nonnegative scoring scale and independent labels. An Elo score is not an appropriate unqualified ratio because its origin is arbitrary. Keep judging competence separate from argument preference.

If unrun: "Planned pilot: replay PeerEval ratings under truthful and rival-downrating profiles, checking rating invariance and measuring selected-argument quality on a fixed nonnegative

preference scale." Replace the orange passage; it is shorter. Do not report a red team's failure to find an attack as a proof against arbitrary coalitions. This pilot ranks below 1 and 3 for the deadline.

C30. Pilot 3: replaying forged outputs - run this second

Location: Task 2.1 preliminary results, p. 12.

Would the expected result convince me? The forgery-acceptance decrease would convincingly establish the value of authenticated execution over trusting a transcript. It does not establish a need for debate or prove Proposition 4's assumptions. If every item comes with a decisive calculator/port query, success may reflect direct verification alone.

Read out from these same 200 cases: what is supplied to each condition, authenticity rejection versus final answer accuracy, remaining failures and tool-call/token cost. Do not add a new predeadline condition: explicitly leave the debate-versus-direct-verification question open if this pilot lacks a tool-using single-judge comparator. Specify that comparison for the already proposed evaluation. Do not describe 0.95 final accuracy as a theoretical prediction.

If unrun: "Planned pilot: on 200 tool-settleable disputes with forged outputs, compare transcript-only judging with exact-call replay, measuring forgery acceptance, decision error and verification cost. This isolates output authentication; benefit beyond a tool-using single judge remains to be tested." Replace the orange passage at a substantial word saving.

C31. Pilot 4: plants and audit flagging - useful only if its two claims are separated

Location: Task 2.2 preliminary results, p. 14.

Would the expected result convince me? AUC 0.56 would show only that the tested discriminator nearly fails on that split. It would not establish indistinguishability to the stronger, differently trained colluder. A median of 40 audited decisions says little about real-stream delay or harm without rho, baseline error, excess-error definition, number of repetitions, and which role is blamed. Zero false flags in an unspecified sample does not validate a 0.05 guarantee.

If unrun: "Planned pilot: compare detectability of templated and controlled-edit plants on held-out source questions, then simulate excess-error flags at specified audit rates. Report AUC uncertainty, audits and total decisions to flag, harm before flagging, and false flags across repeated runs." It replaces the orange passage and is shorter. Keep the formal false-positive guarantee conditional on a valid test, independent of what a small simulation observes.

C32. Pilot 5: RL stability - the most valuable strong result, but too demanding to rely on at the last minute

Location: Task 3 preliminary results, p. 15; eqs. (5)-(6).

Would the expected result convince me? Actual held-out curves showing a durable gain under flawed peer references would support the first-answer reward. A one-run 1.5B/GSM8K result would not establish general stability, message-credit quality, or continued scaling beyond human oversight. Collapsing the baseline at a selected step can reflect hyperparameters; use the matched training budget and checkpoints already in the pilot plan. This pilot does not test eq. (6).

If unrun: "Planned pilot: train a 1.5B model with GRPO using flawed peer references, comparing eq. (5) with majority-agreement and truth-label rewards at matched compute. Held-out learning curves test improvement and collapse; Proposition 6 motivates the conditional reward signal but does not predict final accuracy." This is shorter than the orange passage.

Deadline priority: pilots 1 and 3 give the best evidence per unit time and support RG1/RG2 directly. If pilot 5 is already complete with reliable logs, use its actual curves; do not rush it to manufacture a five-number story. The remaining unrun pilots should be prospective. All replacement sentences here are conditional options, not instructions to add new prose beside the existing passage.

7. Naval relevance and the skim test

C33. The watch-officer story: credible failure mechanism, oversimplified authority and truth

Location: Section 1.1, p. 1; Figure 1, p. 2; Section 1.4, pp. 3-4; testbed B, p. 16.

Copied schedules, conflicting records and a compromised report are a credible analytical scenario. I cannot verify from the supplied material that the specific patrol-tasking workflow reflects an actual Navy procedure. Label it illustrative; do not imply a validated concept of operations or a Navy partner. C12 makes it an officer's time-indexed assessment and ties the output to further verification. In testbed B, be explicit that the database and report tree share a synthetic world state: replay demonstrates correct access to that state, not resistance to a corrupted real-world source. This caveat replaces "built from the same events" in Table 2 with "shared synthetic state; source corruption tested separately" at comparable length.

The basic-research identity is viable: dependence, strategic selection, noisy verification, audit allocation and multi-agent credit recur far beyond shipping. Preserve that breadth through mathematical variables and failure regimes, rather than adding more Navy use-case nouns. No additional naval experiment or new relationship is needed in the remaining day.

C34. Title and three-sentence panel memory

Location: title; Section 1.1, p. 1; Figure 1, p. 2.

The title matches the intended architecture and can stay. "Trustworthy" is defensible as a research objective once the overview states the conditional limits; the draft does not deliver unrestricted trust. "Evaluation" spans finite-answer evidence assessments and narrow executable scientific claims, not a general solution to open-ended peer review. An optional equal-length alternative is "Auditable Collective Reasoning and Evaluation for AI Agents"; the stronger priority is fixing the claims, not changing the title.

Three sentences a busy panelist would carry into discussion: "This project asks how several AI tools can produce a dependable assessment when they copy each other's errors and some are compromised. It combines dependence-aware weighting, impartial role selection, evidence-replaying debate, random audits and learning from peer decisions. The question is relevant, but the overview presents the pipeline as trustworthy before establishing that the learned weights, judge and training references can be trusted under the stated threat model."

This is a skim impression, not proposed insertion; there is no page cost.

C35. Where the non-specialist first gets lost

Location: Section 1.1 F1-F4, p. 1; Figure 1, p. 2; Section 1.2, pp. 2-3.

The first conceptual jump is "verifying every claim is costly, therefore debate": why will generating two arguments and rerunning a call be cheaper than one competent tool user making that call? The first visual overload is Figure 1's detailed group lotteries and agent labels before RG1 is explained. The next is the RG3 overview's "without that agent's group, beyond chance agreement," a technical construction whose purpose is unclear without the noise condition.

Replace the debate-introduction sentence with: "We test whether adversarial debate can expose a small number of decisive checks more cheaply than verifying every claim; where a single tool query already suffices, direct verification is a baseline." This replaces the existing approximately 45-word sentence plus its lead-in at no increase. Apply C9's simplified Figure 1 and replace the RG3 overview's scoring mechanics with "Task 3 tests whether frozen peer references and evidence-sensitive credit improve agents without rewarding unsupported conformity." The shorter overview funds C4's truth-calibration qualification.

8. Calibration against the funded technical proposals

C36. Joshi: a quantified mathematical improvement precedes the new research agenda

Location: Joshi technical PN-6 to PN-8 (PDF pp. 8-10), eqs. (1)-(4), Figure 2; your pp. 7-9, eqs. (2)-(3), Figure 2.

Joshi supplies a baseline MSE, an estimator with a correlation parameter, and an improvement term that varies with that parameter, then identifies new unresolved settings. Your saturation and identifiability statements describe properties of a model but do not yet quantify a new finite-sample decision improvement under learned dependence. **Worth doing within the limit:** yes as a sharper proposed theorem target, not a rushed new theorem or experiment. Replace the broad "rates for detecting groups" promise with a target relating sample size, group separation and calibration/routing error. Use C6/C20's word substitutions. Keep your simulation if verifiable; it should illustrate the conditional setting rather than carry the entire novelty claim.

C37. Joshi and Yousefi: the Navy constraint changes what is optimized

Location: Joshi PN-3 to PN-4 (PDF pp. 5-6); Yousefi technical p. 4 (PDF p. 6), Section 1.3; your Section 1.4, pp. 3-4.

Joshi connects communication limits to compression, training and inference decisions. Yousefi connects unresolved spray physics to specific modeling errors and quantities needed by predictive models. Your naval section mostly names places the framework could be useful. **Worth doing:** yes. Use C12's false/missed departure errors versus verification budget and distinguish stale evidence from forged execution logs. Replace the broad fifth use-case bullet and generic mapping sentence; do not add another transition paragraph or a named collaborator without an actual relationship.

C38. Chen: evaluation variables and denominators are more concrete

Location: Chen Sections 3.3 and 4.7, technical pp. 12-13 and 19 (PDF pp. 15-16 and 22); your Section 3.5, p. 17, Table 4 p. 18.

Chen identifies experimental conditions, counterbalancing, task-time measures and retrieval-performance measures tied to each hypothesis. Your metrics are relevant but often remain nouns without denominators or comparison conditions. **Worth doing:** yes; use C8 to define minority recovery, majority damage, abstention coverage, budget matching and the meaning of scientific-claim truth. Replace the existing metrics paragraph and table cells. There is no need to imitate the human-study design or add a study.

C39. Yousefi and Zhang: completed evidence is distinguished from plans and contingencies

Location: Yousefi Section 3.3.3, technical p. 18 (PDF p. 20), Table 1/Figure 8; Zhang Section 2.3.2, technical pp. 16-18 (PDF pp. 18-20); your preliminary-result passages pp. 9-15.

The examples show concrete dataset/condition information or scenario-specific outcomes with identifiable physical interpretations. Your published prior work is credible feasibility evidence, but the five unrun numeric pilots are not. **Worth doing:** yes, by replacing unrun passages prospectively and giving actual conditions for whichever of pilots 1 and 3 runs. Do not add more figures just to match their visual density. Their domain-specific equation count is not a target for yours.

C40. Chen and Yousefi: contingencies preserve a scientifically interpretable outcome

Location: Chen Section 4.8, technical p. 20 (PDF p. 23); Yousefi Section 3.3.4, technical pp. 18-19 (PDF pp. 20-21); your Task 1.2 risks p. 11 and Task 3 risks p. 16.

Your fallback to prompt diversification does not restore independent model groups, and switching eq. (6) to the debater's own side changes the research objective into a shaped debate-win reward. **Worth doing:** yes, as replacements. Use "no eligible judge -> unsettled" in C7. Replace the final two sentences of Task 3 risks with "If message credit promotes conformity, report that failure and retain the frozen-reference first-answer objective; the debate-win variant is a comparator, not evidence for group-decision credit." This is shorter than the current fallback and keeps a falsifiable result rather than redefining success. Your existing milestone schedule is already adequately concrete; do not overhaul it to imitate the examples.

9. Overlap and compliance

C41. Funded CAREER: limited method overlap, real infrastructure reuse

Location: your Section 4, p. 19; Table 2 B, p. 16. Reference: CAREER PDF pp. 2-3, 6-13, Sections 3.2-3.4.

CAREER develops hierarchical event extraction with question generation/answering, dynamic event-graph construction, consistency rewards and schema induction/matching. A reviewer could reasonably see fresh development of event extraction, question generation or event-graph consistency inside testbed B as already funded. The collective rater model, role-selection mechanism and collusion-audit problem are distinct aims. Your boundary is **partly clear**: you acknowledge reuse, but do not explicitly exclude renewed pipeline research. C11's first sentence fixes this without deleting testbed B. Reward terminology alone is not overlap; the target and charged work matter.

C42. Pending complex-reasoning proposal: moderate benchmark/evaluator overlap

Location: your Tasks 2.1/3 and testbeds B/C, pp. 12, 15-16; Section 4, p. 19. Reference: NSF complex-reasoning PDF pp. 6-9 (Section 3), 11-13 (Section 4), 14-15 (Section 5).

The pending work proposes event-structured multi-hop QA generation, rationale/explanation methods, preference-based explanation evaluation and reward modeling, decomposition into intermediate questions, and hypothesis refinement using checker modules. These are not the same as impartial multi-agent debate or adversarial sampling, but generic "new explanation evaluators," new QA generation and reused checker development could look duplicated. Your boundary is **not explicit enough** because this pending proposal is unnamed in the technical paragraph. Use C11 and specify that already developed extractors/checkers are inputs, while ONR studies their use under dependent and strategic agents. Do not claim that the entire underlying step-verifier training problem is a new ONR aim.

C43. Pending hypothesis proposal: substantial adjacency in execution, rewards and training

Location: your Task 2.1, p. 12; Task 3, p. 15; Table 2 C, p. 16; Section 4, p. 19. Reference: NSF hypothesis PDF pp. 9-12 (Section 4), 12-16 (Section 5).

That proposal trains aspect reward models from scientific reviews and develops ResearchGym: a Dockerized implementation/validation environment, execution tests, reproducibility signals and multi-granularity RL rewards. A reviewer could see building the general code sandbox, executable scientific-claim evaluator or single-agent fine-grained training machinery here as already proposed. Your distinctive ONR scope is **collective** inference and incentives under dependence, unreliable judging and collusion; eq. (6)'s multi-agent credit must earn that distinction. The present boundary is **weak**, not proof of duplicate funding. Apply C11 and describe sandbox reuse as infrastructure, with disjoint scientific aims/effort. I cannot verify cost allocation, NSF submission status, or the Open Philanthropy/Amazon scope from these PDFs; those require actual award/support records, not inference.

C44. Compliance against the FY27 rulebook

Location: technical pp. 1-20; particularly Sections 1.4, 3.6-3.7, 4-6, pp. 3-4, 18-20. Rulebook: Sections C, D and F, printed pp. 12, 15-16, 19-20 and 29-30.

Requirement	Finding within the reviewed scope
20-page technical limit	Sections 1-6 occupy exactly numbered pp. 1-20. Cover and contents are excluded by the NOFO. Reflow after the font fixes is a material risk; no current spare page is available.
Letter size, 1-inch margins, single spacing	Source sets letterpaper and 1-inch margins; PDF is 612 x 792 points. No broad geometry issue found. Font expansion must not be funded by smaller margins.
Times New Roman, 12-point	C9 is an actual risk: scriptsize figures, small tables/captions, Nimbus Roman rather than the named face. The NOFO does not state a figure/table exemption.
Technical approach and alternatives	Present in Sections 2-3. Closest-method comparison and mathematical scope need C4-C8.
Primary ONR S&T Focus Area	Present on technical p. 3: Artificial Intelligence and Autonomy; associated Intelligent and Collaborative Agents and Adversarial AI & Autonomy match the NOFO list. No requirement to add a secondary area.
Future naval relevance	Present, pp. 3-4. C12 strengthens the content; the section itself is not missing.
Schedule/milestones; goals/benchmarks	Present in Tables 3-4 and Section 3.6, pp. 17-18. Replace vague criteria, not the whole schedule.
Gold Standard Science	Present in Section 3.7, pp. 18-19. Future commitments are specific; prior success is asserted too generally. See C45.
Management, single PI, qualifications, responsibility	Present, pp. 19-20. Six responsibility headings track the required topics. Factual claims about citizenship/residency, commitments and institutional systems are not independently verified here.
References/CV/letters/DMP/support documents	Excluded from this technical-only review; no completeness or compliance certification is made about material after numbered p. 20 or other application uploads.

Do not import older examples' omissions. The named focus area and Gold Standard Science section are newer required content; keep both. The NOFO's five selection criteria are equally important; this requested technical-only verdict is not a prediction of the final award decision.

C45. Gold Standard Science: demonstrate prior practice with a checkable example

Location: Section 3.7, pp. 18-19; documented prior resources in Section 5, p. 20.

The NOFO asks to "Describe your success in implementing Gold Standard Science," not only future intentions. The sentence "as the PI's previous work does" is too generic. **Replace the opening and the trailing generic reproducibility assertion with:** "The PI's public PeerEval leaderboard and MEQA benchmark provide prior examples of releasing evaluation resources. This project will extend that practice through versioned logs, held-out evaluation, stated assumptions and publication of failures." This uses resources the proposal already identifies, rather than inventing an unverified historical protocol. Pay for it by replacing the existing opening sentence, removing "and our group studies reproducibility" and its citation, and cutting the later generic clause "with results for peer-reviewed venues in each." If public availability cannot be substantiated, use a verified release instead. Do not claim a prior preregistration or statistical practice not documented.

The section broadly covers the order's tenets, but "decisions carry calibrated probabilities" needs the conditional wording in C4. A random audit does not automatically certify calibration, and a confidence interval requires the appropriate sampling assumptions. Replace that phrase with "decisions carry probabilities whose calibration is checked on held-out data"; pay the added words by deleting the redundant "with valid confidence intervals" assertion until the estimator is specified.

C46. Submission-readiness actions within the technical file

Location: all five preliminary-result passages; Section 3.1 R5, p. 6; final source build.

Replace all orange hypothetical numbers or label the work prospectively as C28-C32 specifies; then enable the final-build guard. Also change R5 from "so agents cannot learn to spot them" to "so policy updates do not directly train on planted items; recognition remains possible." Its existing word count funds this correction. Exclusion from training does not eliminate in-context learning, distributional recognition or adaptation from rewards/flags.

Keep the 20-page limit after recompilation. Search the submitted technical PDF for "pilot," "expected," "pending" and every now-removed numeric placeholder, visually inspect all pages at the intended print size, and verify that references are in the intended separate upload. These are document checks, not experiments. Deadline in the supplied NOFO is 30 October 2026, 5:00 p.m. Eastern, which is **2:00 p.m. Pacific**. That timing is not a review of the excluded application forms.

10. Fifteen sentence-level rewrites

These are local replacements, mostly on pp. 1-5. They overlap the earlier fixes; use them as alternatives, not additional paragraphs. Where a revised sentence is longer, its named cut pays for the difference.

C47. Central problem: finite claims and resources

Location: Section 1.1, p. 1, bold central question.

Original: "The central problem of this project is how a group of AI agents of unknown and correlated reliability, some possibly dishonest, can reach trustworthy decisions and evaluations."

Revised: "The central question is when agents of unknown, correlated reliability, including strategic agents, can produce accurate, auditable assessments under limited verification."

Payment: shorter replacement; it also aligns the question with conditional basic research.

C48. The operational consequence

Location: Section 1.1, p. 1, vessel paragraph.

Original: "A majority vote picks the wrong answer and sends the patrol to sea after a ship still at berth; open discussion may do no better, because the compromised agent is persuasive and the two agents with the same source back each other up."

Revised: "A majority vote recommends acting on a false departure assessment; open discussion may preserve the error when copied reports and a persuasive compromised agent reinforce one another."

Payment: shorter. Use either this sentence or C12's paragraph replacement.

C49. Why debate rather than a direct check?

Location: Section 1.1, p. 1, debate-introduction sentence.

Original: "We propose debate as a foundation for trustworthy evaluation by AI: two agents argue for opposite answers, exposing each other's flaws, until the dispute narrows to one step a judge can verify."

Revised: "We test whether opposing arguments expose a few decisive checks more cheaply than verifying every claim, using a tool-using single judge as a baseline."

Payment: shorter. Retain the existing debate citation if the original explanatory sentence is removed.

C50. Failure F1: avoid an unnecessary trend generalization

Location: Section 1.1 F1, p. 1.

Original: "Model errors are correlated, most of all among larger, more accurate models ..."

Revised: "Shared model errors make agreement weaker evidence than the number of votes suggests ..."

Payment: same length or shorter; keep only references that support the revised claim. This removes an unnecessary universal scaling suggestion.

C51. Failure F4: keep the distinction between a reported result and a universal law

Location: Section 1.1 F4, p. 1.

Original: "Agents trained to match the group's answer are rewarded for conforming and can learn to agree on mistakes: training a model on its own majority answers ended in a sudden collapse of accuracy."

Revised: "Agreement rewards can reinforce shared mistakes; prolonged majority-based self-training has produced sudden accuracy collapse."

Payment: shorter. Retain the citation to Can Large Reasoning Models Self-Train?, which reports both early improvements and later collapse.

C52. Broader motivation: remove the speculative inevitability

Location: Section 1.1 continuation, p. 2.

Original: "As AI systems surpass people at more tasks, people cannot easily evaluate the systems' answers or their debates, so oversight will rely on AI evaluating AI, with people auditing a small part of the work."

Revised: "As evaluation becomes harder, AI-assisted oversight becomes useful only if limited trusted checks can expose systematic and strategic failures."

Payment: shorter; this makes the research question visible instead of assuming the proposed solution scales.

C53. RG1 overview: bound the claim to its model

Location: Section 1.2 RG1, p. 2.

Original: "Task 1.1 weighs each answer by its agent's credibility, estimated without an answer key, so that a group sharing a mistake counts at most once ..."

Revised: "Task 1.1 estimates dependence-aware answer weights from unlabeled responses, uses separate calibration questions, and bounds a group's influence under the stated model ..."

Payment: delete the exam analogy in Task 1.1; this is an alternative overview formulation to C4's wording.

C54. RG3 overview: remove the unsupported scaling consequence

Location: Section 1.2 RG3 continuation, p. 3.

Original: "When that reference is more often right than wrong, training follows the agents' true accuracy in expectation, without human labels, so it can keep working after the agents outgrow human judges."

Revised: "For a frozen binary reference with conditionally independent errors below one half, a per-question gradient identity motivates training; shared errors and changing references remain explicit risks."

Payment: shorter; the qualifier is essential, not optional technical detail.

C55. Hypothesis: state a testable comparison

Location: Section 1.2, p. 3, central hypothesis.

Original: "... learns from its decisions through rewards that agreement alone cannot raise."

Revised: "... tests whether peer-reference and message rewards improve held-out accuracy without increasing shared errors or attack success."

Payment: cut the next generic sentence "Each task controls one source of error ..." (the framework explains this again). This also funds reporting the negative outcomes.

C56. Mathematical confidence: distinguish foundations from proposed theory

Location: Section 1.3, p. 3, "Why it will succeed."

Original: "Each method follows from a principle whose property we prove for agents that share mistakes, argue, and may collude ..."

Revised: "Existing results justify idealized components; the proposed theory studies their finite-sample and adversarial limits ..."

Payment: shorter. Use this or C6's longer two-sentence replacement, not both.

C57. Naval mapping: copied and stale are different

Location: Section 1.4, p. 4.

Original: "Here, Task 1.1 counts copied and stale reports once ..."

Revised: "Task 1.1 discounts dependent reports; Task 2.1 checks whether reproduced evidence is current and supports the disputed claim ..."

Payment: cut the later clause "and each decision states its reasons," which the paragraph and figure already establish. Staleness is not fixed merely by counting a report once.

C58. Dependence literature: credit the closest model

Location: Section 2, p. 4, reliability paragraph.

Original: "... and later work adds ... dependence between classifiers (Jaffe et al., 2016) ..."

Revised: "... and later work includes the latent-group dependent-classifier hierarchy we adapt (Jaffe et al., 2016) ..."

Payment: shorten the adjacent list of implementation history; no additional paragraph. Use the direct paper link when checking this relationship.

C59. State-of-the-art gap: avoid a brittle universal negative

Location: Section 2, p. 5, debate paragraph.

Original: "... but no judge re-runs the agents' own calls. We know of no debate protocol designed for many agents that share mistakes or collude."

Revised: "Our comparison isolates exact-call replay, dependence-aware routing, impartial roles and adversarial audits, rather than treating tools or multi-agent debate alone as new."

Payment: shorter. The search verifies close antecedents; it cannot certify that no existing system does replay.

C60. Preliminary simulation and identifiability

Location: Task 1.1, p. 9, preliminary-results opening (one of the later-page exceptions).

Original: "The simulation in Figure 2(a) bears out Propositions 1 and 2."

Revised: "Figure 2(a) illustrates finite-sample behavior under the assumed model; it does not establish identifiability or robustness outside that model."

Payment: cut the p. 7 repeated explanation of the same simulation's curves, as in C10. A simulation can illustrate a theorem, not validate population identifiability.

C61. Potential-based credit

Location: Task 3, p. 15, after eq. (6) (later-page exception).

Original: "... and credits each message with its effect given the earlier ones ..."

Revised: "... and assigns order-dependent confidence increments; these are not causal effects of individual messages ..."

Payment: same approximate length. Keep the existing statement that policy invariance does not carry over and use C5's caveat instead of adding a separate defense.

Rulebook wording and novelty-check record

C62. Exact review criterion and implications for this review

Location: FY27 NOFO Section F.2, printed pp. 29-30; proposal Sections 1.3, 1.4, 3 and 5, pp. 3-20.

The technical-merit criterion reads:

> (1) Technical Merit - A creative research proposal, demonstrating the potential for making progress in an ONR research area and the degree of innovation, and soundness of technical concept. The Principal Investigator's awareness of the state of the art and understanding of the scope of the problem, significance and originality of the technical approach and effort needed to address/solve the problem and anticipated scientific impact within the field. The Principal Investigator's capabilities, related experience, facilities, techniques or unique combinations of these which are integral factors for achieving the proposal objectives; and the qualifications, capabilities and experience of key personnel and team members who are critical to achieving the proposal objectives. (Note: this is single PI award, no co-PI's allowed).

The scientific/alignment portion of criterion (2) reads (including its punctuation):

> (2) Naval Science and Technology Priorities - Basic and Applied Research that address new knowledge to develop new technologies that ultimately become new capabilities delivered by the acquisition community (as described in the ONR Science and Technology Department section of ONR's website at https://www.nre.navy.mil/organization/departments. And the alignment to an ONR S&T Focus Area (see section C. Program Description).

The remainder of criterion (2) addresses research-security risk assessment. This review does not infer a risk from nationality or evaluate excluded disclosure forms. The NOFO states, "Criteria 1, 2, 3, 4 and 5 are equally important." The exact formatting rule is "Font - Times New Roman, 12-point"; the technical limit is 20 pages, with the specified exclusions. These quotations come from the supplied annotated rulebook; annotations do not create a font exception.

C63. Verified novelty boundaries, with direct paper links

Location: Section 2, pp. 4-5; Sections 3.2-3.4, pp. 7-16; Table 1 p. 6.

The following are the decision-relevant literature checks, not a request to add another related-work page. Most are already cited in the source. Replace generic history and overbroad novelty claims with precise boundaries.

- Unsupervised Ensemble Learning with Dependent Classifiers (Jaffe et al., 2016): the closest model predecessor; C6 states the relationship. Direct attribution is required next to the model.
- Dependence-Aware Label Aggregation for LLM-as-a-Judge via Ising Models (Balasubramanian et al., 2026): dependence-aware aggregation already supplies separation results and posterior modeling. A claim that shared dependence has not been modeled is untenable.
- CARE: Confounder-Aware Aggregation for Reliable LLM Evaluation (Zhao et al., 2026): already studies shared confounders with identifiability and finite-sample recovery. The new target must specify the debate/strategic setting and its added assumptions.
- Tool-MAD (Jeong et al., 2026) and DynaDebate (Li et al., 2026): tools, step critiques and a verification agent are not individually new. DynaDebate's verifier chooses and executes a new tool command on disagreement; I did not find an exact replay requirement in the reviewed method text. This is a comparator distinction, not proof of priority for replay.
- Truthfulness Despite Weak Supervision: Evaluating and Training LLMs Using Peer Prediction (Qiu et al., 2026): peer-derived label-free post-training is already established as a research direction. Its appendix explicitly examines collusive participant composition and reports failures in some regimes; do not imply prior work simply ignores collusion. Robustness guarantees against the proposal's particular strategic agents remain a legitimate gap.
- Co-RL: Unsupervised Reasoning Emerges from Diverse Cohort in Multi-agent RL (Yang et al., 2026): a direct peer/cohort training comparator. I verified the abstract and peer-derived-reward/diversity claims, not every implementation detail of its loss.
- Debate Training Reduces Reward Hacking in RLAIF (Kenton et al., 2026): debate training and its judge-hacking limitations are close prior art; the proposed learning objective must earn its advantage over this alternative rather than compare only to an unspecified debate-win reward.
- AI Safety via Debate (Irving et al., 2018), Scalable AI Safety via Doubly-Efficient Debate (Brown-Cohen et al., 2024), and Informed Truthfulness in Multi-Task Peer Prediction (Shnayder et al., 2016): the theoretical lineage is appropriately present; its ideal guarantees cannot be inherited without the corresponding assumptions.

- Avoiding Imposters and Delinquents (Steinhardt et al., 2016): fixed trusted work in a batch curation model is not the online per-attack guarantee claimed here.

Search limit: I verified the closest structural overlap and contemporary primary sources; I did not establish exhaustive priority, independently reproduce your published results, or verify every quantitative literature claim on p. 1. No claim in this review treats a failed search as evidence that a method is new. The defensible novelty is in explicitly specified new statistical/adversarial questions and their interaction, not the existence of weighting, debate, audits, replay, or peer rewards individually.

C64. One-day editing order and a page-neutral budget

Location: entire technical narrative, pp. 1-20.

1. Fix C3-C7 and remove every unrun numerical claim. These are claim/assumption corrections, not new research.
2. Run pilots 1 and 3 if feasible from existing code; otherwise use the prospective replacements. Do not hold the submission for pilot 5.
3. Delete Table 1 and Figure 5, shorten Figure 1, and reflow all ordinary figure/table/caption text at the required size. This space is reserved for compliance, not an extra methods section.
4. Replace the existing evaluation paragraph/table cells and support-boundary paragraph. Check award scopes for the latter.
5. Apply only the nonduplicative early-page line edits, enable the final build, recount numbered pages, and visually inspect the submitted PDF.

Every scientific addition above has a specific replacement/cut. The large graphic cuts fund larger fonts; they should not be double-counted as available prose space. If reflow still exceeds 20 pages, shorten repeated qualifications/service material on pp. 19-20 before cutting the assumptions, evaluation definitions, or Gold Standard Science section.